

Caroline Dakoure

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<https://carodak.github.io> | <https://github.com/carodak/>

Profile

- ∴ Strong interest in the ethical use of new technologies, games, XR, with a positive impact on humans and their environment
- ∴ Strong communication skills, especially for collaboration and mediation
- ∴ Analytical mindset, conscientious, and attention to detail
- ∴ Languages: French and English

Education

Master's Degree in Computer Science (Artificial Intelligence) Adaptive VR Emotional Environments Under the direction of Dr. Claude Frasson, University of Montreal	2020
Bachelor's Degree in Computer Science Université de Montpellier, France	2017
Scientific High-School Diploma (specialization in Mathematics) Lycée Jacques Prévert, Saint-Christol-lès-Alès, France	2014

Specific knowledge

Game Engine: Unity	Programming Languages: C++, C#, C, Java, Python, Scheme, Bash
DevOps Automation Tools: Docker	Machine Learning: Librairies Scikit-learn, Numpy, TensorFlow, Keras, PyTorch, etc.
Data Management: SQL, SGBD MySQL, Castor, Redcap	Web: PHP, React, Javascript, HTML5, CSS3,

Work experience

Research Software Developer , Centre de Recherche Douglas, Québec	Nov. 2020-Now
∴ Lead VR application development: MRI Simulation, Lobby Room (Unity, C# with design patterns), from requirements analysis to production; design, development, testing, optimization, and Meta Store deployment ∴ Conceptualize and lead development of web applications using React, PHP, JS, HTML, CSS (e.g., a match-3 puzzle game with procedural content generation, and an e-learning platform) ∴ Analyze XR (AR/VR) requirements and learn new computer techniques to address them ∴ Develop cognitive tests in Python and JavaScript and create Python scripts to automate tasks ∴ Supervise, mentor, and share expertise with new members of the programming team ∴ Provide recommendations on computer system-related issues	
Student trainee in a Computer Science and Research Laboratory LIRMM , Montpellier, France	Jul. 2017
∴ Experiment deep learning (Keras, TensorFlow, Scikit-learn) and node2vec (with Networkx) algorithms to learn new representations of the features and get better predictions ∴ Implement an architecture in Python with several classifiers (neural network, decision tree) to determine graph properties (e.g., planarity, and tree structures)	

Student projects and personal achievements

VR environments that change according to the patient's emotions as part of my master's degree project at UdeM

2020

- ∴ Develop adaptive [VR environments](#) using Unity3D and C#
- ∴ Create an NPC pathfinding & Goal-Oriented Action Planning system.
- ∴ Coordinate paired programming to develop a [ML classifier for emotion detection](#)
- ∴ Plan tasks to meet company deadlines & participate in team meetings
- ∴ Find strong scientific theories to choose appropriate environments

Achievements: Created two real-time changing virtual reality (VR) environments using Unity3D based on patients' emotions. Developed a machine learning model using Python to detect the intensity of confusion (ranging from 0 to 1) from EEG signals.

Development of procedural content as part of a course on modern computer games at McGill

2018

- ∴ Simulate the behavior of various types of agents (steering behaviors, collision detection, procedural content generation, game physics)
- ∴ Develop a system where movement is based on directional forces

Achievements: Created a simulation in Unity3D and C# featuring three different types of agents. Some agents needed to reach an exit while others interfered or socialized.

Development of a Gomoku game as part of a second-year bachelor's project

2016

- ∴ Create [a game in C++](#) where 2 players can play Gomoku
- ∴ Implement the MinMax algorithm to allow a player to play against an AI
- ∴ Enhance the MinMax algorithm using the alpha-beta pruning technique
- ∴ Implement a specific variation of Gomoku (Renju)

Used the Code::Blocks IDE with the SFML library (C++) for development

Publications

<https://orcid.org/0000-0002-3195-3643>

- Co-author, CCGH 2023.
- First author in FLAIRS 2021 Conference Proceedings (34) and BFAL 2020, Springer

Other Achievements

Participant, MIT Reality Hack 2025

2025

- ∴ Over 5days, we developed [a musical game in MR](#) using Unity

1st price, Winner of the NeuroVR Hackathon 2024 organized by McGillXR

2024

- ∴ Over 24hours, we developed [the first version of an MRI simulation](#) in VR using Unity

1st price in a Competition, Winner of the 2019 Serene-Risc CyberStats Challenge

2019

- ∴ Creation of [an informative poster](#) to promote awareness messages on Cybersecurity and Cybercrime using data from Statistics Canada's and presentation at GoSecure 2019

Social implications

Mentor during McGillXR Hackathon (2025), **Bikes Upcycling** for Togo at CycloNordSud (2023), **Animal Care** at Café Chato (2022), and **Participant Welcoming and Registration** for "Ça va aller" campaign event at UdeM (2019)